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Special Communication

Low-cost algorithms for clinical notes phenotype classification to enhance epidemiological surveillance: A case study

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ABSTRACT

Objective: Our study aims to enhance epidemic intelligence through event-based surveillance in an emerging pandemic context. We classified electronic health records (EHRs) from La Rioja, Argentina, focusing on predicting COVID-19-related categories in a scenario with limited disease knowledge, evolving symptoms, non-standardized coding practices, and restricted training data due to privacy issues.
Methods: Using natural language processing techniques, we developed rapid, cost-effective methods suitable for implementation with limited resources. We annotated a corpus for training and testing classification models, ranging from simple logistic regression to more complex fine-tuned transformers.
Results: The transformer-based, Spanish-adapted models BETO Clínico and RoBERTa Clínico, further pre-trained with an unannotated portion of our corpus, were the best-performing models (F1= 88.13% and 87.01%). A simple logistic regression (LR) model ranked third (F1=85.09%), outperforming more complex models like XGBoost and BiLSTM. Data classified as COVID-confirmed using LR and BETO Clínico exhibit stronger time-series Pearson correlation with official COVID-19 case counts from the National Health Surveillance System (SNVS 2.0) in La Rioja province compared to the correlations observed between the International Code of Diseases (ICD-10) codes and the SNVS 2.0 data (0.840, 0.873, and 0.663, p-values $\leq 3 \times 10^{-7}$). Both models have a good Pearson correlation with ICD-10 codes assigned to the clinical notes for confirmed (0.940 and 0.902) and for suspected cases (0.960 and 0.954), p-values $\leq 1.7 \times 10^{-18}$.
Conclusion: This study shows that simple, resource-efficient methods can achieve results comparable to complex approaches. BETO Clínico and LR strongly correlate with official data, revealing uncoded confirmed cases at the pandemic's onset. Our results suggest that annotating a smaller set of EHRs and training a simple model may be more cost-effective than manual coding. This points to potentially efficient strategies in public health emergencies, particularly in resource-limited settings, and provides valuable insights for future epidemic response efforts.

Abbreviations: AdamW, Adaptive Moment Estimation with Weight regularization; ANOVA, Analysis of Variance; BERT, Bidirectional Encoder Representations from Transformers; BETO, Spanish Pre-Trained BERT Model; BiLSTM, Bidirectional Long Short-Term Memory; BOW, Bag of Words; EHR, Electronic Health Record; ICD-10, International Classification of Diseases, 10th revision; GDPR, General Data Protection Regulation; GloVe, Global Vectors for Word Representation; GPUs, Graphics Processing Units; HIPAA, Health Insurance Portability and Accountability Act; IAA, inter annotator agreement; LR, logistic regression; MAMA, Model-Annotate-Model-Annotate cycle; ML, Machine learning; NER, Named Entity Recognition; NLP, Natural Language Processing; PCR, polymerase chain reaction; SNVS 2.0, Version 2 of the *Sistema Nacional de Vigilancia de Salud* (Argentina's National Health Surveillance System); TAPT, Task-Adaptive PreTraining; TF–IDF, Term frequency–Inverse document frequency; RawCRC, Raw Clinical Reports Corpus; RoBERTa, A Robustly Optimized BERT Pretraining Approach; WHO, World Health Organization; XGBoost, eXtreme Gradient Boosting

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1. Introduction

In the field of epidemic intelligence, systematic information analysis is essential for detecting, verifying, evaluating, and investigating health events and risks with the aim of early warning. This process relies on two complementary surveillance strategies: event-based and indicator-based surveillance. Electronic Health Records (EHRs) serve as valuable sources of information for event-based surveillance strategies [1].

The emergence of the COVID-19 pandemic in 2020 presented significant challenges, characterized by uncertainty about the limited understanding of the virus's transmission dynamics and associated symptoms. In response, national and international organizations developed and updated recommendations and surveillance strategies based on the best available information on the virus. The World Health Organization (WHO) established emergency ICD-10 (International Classification of Diseases, 10th Revision)[2] codes for COVID-19 infection in February 2020, when virus cases were already documented in four continents. This system standardizes diagnostic codes globally.

Identifying diagnoses from EHRs usually involves manually reviewing the EHR texts and assigning standardized codes to them. Implementing a new ICD-10 code during the pandemic resulted in a lack of standardized coding for some time, exacerbating the complexities of identifying and analyzing epidemiological trends. This situation potentially delayed crucial public health interventions and resource allocation. During the COVID-19 pandemic, healthcare providers faced challenges in accurately documenting and reporting cases, particularly in resource-constrained settings where PCR and antigen tests were scarce due to cost and limited availability of reagents. This scarcity disproportionately affected low-resource countries compared to their high-resource counterparts, highlighting significant disparities in access to diagnostic tools. Importantly, all these considerations remain relevant after COVID-19, as two new epidemics have emerged since then: monkeypox and Oropouche fever.

In response to the pressing need for timely surveillance at that time, we pursued the automatic classification of unstructured data from electronic health records from La Rioja, Argentina, into five categories: no-COVID, COVID-confirmed, COVID-suspected, COVID-rejected, and insufficient information. Automating this process through the use of natural language processing (NLP) techniques offers several advantages, including the ability to generate more timely statistics than indicator-based surveillance, thus complementing data provided by the National Health Surveillance System (SNVS 2.0 from its initials in Spanish). Early onset detection allows the allocation of skilled personnel to other critical tasks other than manual classification, and potentially anticipates a pandemic not yet classified in the ICD-10 system.

There are several challenges in applying NLP to EHRs, including dataset imbalance, the scarcity of publicly available corpora, particularly in Spanish – mainly due to the sensitivity of medical information, and the need to safeguard patient privacy –, as well as the abundance and ambiguity of abbreviations in clinical notes [2–6]. An additional challenge is the constitution of an interdisciplinary team to address solutions for this problem [4]. Although this offers several advantages, achieving a common understanding among different areas of knowledge requires considerable effort.

Previous studies have shown that employing simple, interpretable approaches with rapid deployment capabilities can yield promising results for this task, although such insights are predominantly available for clinical notes in English [2,7,8]. Furthermore, in many cases, comprehensive descriptions of the methodologies employed and their performance metrics are lacking. Some works also discuss the importance of automating ICD-10 coding [9,10]. To bridge these gaps, this work offers a detailed description and evaluation of machine-learning solutions to classify clinical notes concerning COVID-related categories to provide decision-makers in Spanish-speaking contexts with vital information.

Given the time and resource constraints, particularly prevalent in low- and middle-income countries, as well as the urgency of a pandemic, our approach prioritized rapid and cost-effective methodologies. We focused on factors such as the availability of pre-trained models and the cost associated with training them, including both human resources and electricity consumption [11,12]. Additionally, the absence of high-cost resources such as GPUs (Graphics Processing Units) further constrained our approach. Our study evaluates a range of algorithms, including Naïve Bayes; Logistic Regression; Gradient Boosting Classifier (XGBoost); a bidirectional recurrent neural network (BiLSTM); and two transformer-based approaches: BETO Clínico and RoBERTa Clínico, which we developed by further pre-training existing models with an unannotated dataset derived from our corpus. For training and evaluating our algorithms, we required annotated data. The limited availability of annotated resources, the shortage of specialized annotators, and the minimal resources allocated for annotation posed additional challenges.

Given these challenges, our study aims to assess whether accurate and actionable classification results can be achieved at a lower cost than with state-of-the-art approaches. Our findings reveal promising results achieved within a short timeframe. Our best-performing model was BETO Clínico (a state-of-the-art transformer model based on BETO [13], which is an adaptation to Spanish of BERT, a pre-trained deep learning model that understands the context of words in a sentence [14]). Notably, however, the simpler and more resource-efficient method of Logistic Regression showed a performance comparable to that state-of-the-art approach. Both methods correlated well with case confirmations of the province under study, provided by the National Health Surveillance System, and with ICD-10 codes, when available. Furthermore, Logistic Regression offers the advantage of being more accessible to a wider pool of personnel, facilitating implementation in resource-constrained settings.

Furthermore, our results show that in our problem the automatic coding of EHRs into ICD-10 codes, a labor-intensive task, could be achieved by annotating only a portion of the complete dataset and then training a simple machine learning model to classify the rest. However, it is worth noting that an expert should review the classification results afterwards.

This study is significant because it addresses the urgent need for low-cost, efficient machine learning (ML) methods that can operate effectively in resource-constrained environments, such as those found in many low- and middle-income countries, especially in light of the recent emergence of epidemics, and provides solutions to enhance decision making in these contexts. It also focuses on Spanish EHRs, underrepresented in biomedical NLP. By guiding improvements in pandemic preparedness and response across global settings, we aim to enhance the utility and accessibility of automated surveillance solutions within the broader context of epidemic intelligence.

[2] ICD-10: https://icd.who.int/browse10/2010/en

Statement of significance

Problem	The COVID-19 pandemic highlighted the critical importance of timely, automated systems for detecting health risks in regions where resources, including timely access to trained personnel, are limited.
What is Already Known	Although there have been prior studies on the classification of unstructured texts from EHRs, they are mainly for English. Some lack detailed methodology descriptions and provide only aggregate performance metrics.
What this Paper Adds	Our work provides resource-efficient, scalable methods with state-of-the-art results to help public health authorities identify cases more quickly and accurately, even in the absence of standardized coding systems and with limited annotated data. This study addresses Spanish EHRs, underrepresented in biomedical NLP despite being the second most spoken language by native speakers [15].

2. Related work

The worldwide increasing adoption of Electronic Health Records has transformed them into a valuable source of epidemiological data. Strategies for processing these data, such as information extraction and classification, have been accelerating [7], as it can improve the availability of information for healthcare management [16,17], while real-time access ensures timely and accurate decision-making [3]. EHRs contain structured and unstructured data (composed of clinical notes and discharge summaries, among others). The classification and information extraction from unstructured data is achieved using natural language processing techniques.

As shown by an extensive survey of approximately 150 research studies focused on addressing various aspects of the COVID-19 pandemic in English datasets [2], extraction of information from clinical notes is more difficult and less developed than retrieval of information in biomedical literature, sentiment analysis in social media, and detection of public health issues based on news articles, mainly due to the lack of publicly available corpora, grammatical and spelling errors, and inconsistent usage of language.

Aliabadi et al. [3] also highlight some of the challenges in using EHRs to improve disease surveillance systems, including low quality of data within EHRs, difficulty in extracting information from unstructured data, concerns around data privacy, and multi-disciplinary work needed to address these solutions, also mentioned in [4]. Other studies highlight among the challenges of doing NLP in clinical texts the scarcity of publicly available annotated data and the high cost of annotating it, due to the difficulty of having annotators with medical expertise, defining annotation criteria, and obtaining reasonable inter-annotator agreements, the highly imbalanced data, which is challenging for machine learning algorithms, and the importance of interpretability of the methods [4,5]. Additionally, data has to be anonymized, and both national and international regulations, such as the Health Insurance Portability and Accountability Act (HIPAA) of the United States and the European Union's General Data Protection Regulation (GDPR), must be taken into account to protect patients' data privacy [6].

As mentioned previously, the techniques for automatic classification of EHRs require annotated data. Since timely responses were needed in all cases, an effort was put only in a few cases into developing new techniques or annotating a large corpus with domain experts [2]. Given the sensitive nature of medical data, the difficulty of its annotation process [8], and the novelty of the virus and the disease, at that time there were no Spanish annotated data publicly available.[3] Moreover, most existing tools to process clinical text are developed for English (eg. [2,5,19,20]) or were conducted in English-speaking countries [6, 21].

There are different NLP methods that can be used to automatically extract information from EHRs. Classification and named entity recognition (NER) are two of them [7]. Classification has been used for different tasks, such as automatic assignment of ICD-9 codes to clinical records [9,10,22], automatic classification of the medical subdomain of a clinical note, such as cardiology or neurology [23], EHR phenotyping[4] [21], and other tasks [24,25]. Among the studies using classification to detect the presence or absence of COVID, some fast-deployment NLP tools also consider the importance of simplicity like ours—they, however, were built for English, were evaluated with a very reduced annotated dataset [8,26], and lack a detailed comparison of complex versus simple methods. In other studies, no details about the machine learning methods are provided, and only aggregate performance metrics are available [27]. Still, other studies use NER to detect COVID-19 symptoms and sometimes modifiers (e.g. negations) in English EHRs [20,28–30] or predict test results based on patients' self-reported symptoms [31]. To our knowledge, no such study has been published for Spanish.

In their article [6], Hossain et al. mention that ICD-9 code classification is one of the current trends in medical NLP research. Among the different challenges of applying NLP to EHRs, they mention the imbalance in the datasets, the scarcity of medical data, the prevalence and complexity of medical abbreviations, and the sensitivity of medical data, which requires the protection of patient privacy. In their paper, they compare machine learning models in terms of advantages and disadvantages, focusing on computational efficiency, ease of implementation, ability to handle different types of data (structured, sequential, large volumes), and robustness to issues such as overfitting or noise. They also highlight limitations in scalability, resource requirements, and challenges in the interpretability of results. In particular, they mention that Naïve Bayes and logistic regression are easy and fast to implement, but Naïve Bayes provides slightly lower accuracy, while logistic regression can overfit in high-dimensional spaces. XGBoost is efficient in terms of computation times, but it does not perform well on sparse and unstructured data. BiLSTM is useful for unstructured data, but it is computationally expensive. Regarding feature extraction methods for EHR, they highlight that traditional techniques like TF–IDF, BOW, and GloVe were used more frequently than newer approaches like FastText and BERT. Finally, they mention that most papers on ICD-9 code classification did not focus on machine learning techniques. Finally, Yang et al. mention in the methodological review they performed for ML approaches for EHR phenotyping [21] that only a few articles evaluated external validity or used multi-institution data, and that code is usually not released.

In the context of Spanish, several challenges have previously been organized for information extraction in the medical domain. For instance, CodiEsp [32] focused on assigning tags at a document level, among other tasks. Other initiatives aimed at anonymization [33], and NER, negation detection and relation extraction [34–36].

3. Methods

In this section, we describe the data selection, the annotation process performed, a brief data statement, and the different approaches

[3] There is a publicly available corpus for Spanish, that contains X-ray and computed tomography images, radiological reports with their findings and locations, and PCR results, among others, but no clinical notes [18].

[4] Defined in [21] as "a procedure that uses EHR data to assert characterizations about patients".

developed for doing clinical notes classification on a dataset of electronic clinical records from La Rioja, Argentina. We also present the experimental settings.

For an analysis of potential biases in the predictive algorithms, including possible fairness criteria to detect bias in the classifiers, an analysis of sensitive attributes, and a demographic distribution analysis of the database in relation to the population of La Rioja province, refer to Appendix C.

The annotation criteria, annotated corpus and prototype Natural Language Processing classification tools were developed between March and June 2021.

3.1. Dataset selection

The analyzed dataset consists of 636,543 de-identified clinical reports of medical visits associated with textual descriptions and metadata, like medical specialty and ICD-10 diagnostic codes. From this dataset, we excluded nursing entries (since suspected COVID cases always involve further procedures beyond nurse care) and only considered records from January 1st, 2020 to January 28th, 2021. Only 0.56% of the remaining reports mention "COVID" or at least one of its primary symptoms; 0.44% have ICD-10 codes related to COVID (U071 confirmed and U072 suspected; 1 184 and 1612, respectively); 26.43% have a diagnosis other than COVID; and 72.58% belong to unrelated specialties and do not mention COVID.

The imbalance between COVID and non-COVID in the available clinical notes made the selection of examples for annotation non-trivial. We worked on selecting a dataset to be annotated, as similar as possible to the target population to be predicted, but also containing enough examples of each class. We selected varied examples, covering many population subgroups, maintaining a balanced ratio between negative and positive examples.

Since ICD-10 codes were inconsistently used (because they were not available at the start of the pandemic [37,38] and the criteria for diagnosing COVID changed over time, they were not reliable to train an automatic classifier for detection of COVID cases. Therefore, we decided to construct a manually annotated dataset.

3.2. Manually annotated corpus

An annotation schema and annotation guidelines were developed by two natural language processing researchers with experience in annotation of biomedical texts and two epidemiology residents. Therefore, the regulations published by the Argentinian National Ministry of Health that contained the operational definitions of a COVID case, detailing signs, symptoms, and other relevant epidemiological aspects, were used [39,40].

The labels selected for the problem were *COVID-suspected*, *COVID-confirmed*, *COVID-rejected*, *no-COVID*, and *insufficient information*. COVID-confirmed corresponds to cases where there was a positive PCR or antigen test result, or where symptoms indicated a positive case, or where the professional described the case as positive. COVID-rejected corresponds to cases where the presence or absence of COVID-19 was determined based on the results of PCR or antigen tests. COVID-suspected refers to cases where COVID-19 is suspected, either based on symptoms reported by the patient or when the treating physician diagnoses the case as suspected, even in the absence of reported symptoms. no-COVID refers to cases where COVID-19 is not present, including instances where COVID is not mentioned or it is referenced generally without relation to the patient. Insufficient information refers to cases where the available information is ambiguous or insufficient to determine any of the previous labels.

Following the guidelines outlined by Bender and Batya [41], we present a summary of the Data Statements. The annotation process involved three domain expert annotators (advanced medicine students) and two coordinators (the previously mentioned epidemiology residents). They all are native Rioplatense Spanish speakers familiar with reading, in the first case, and also writing, in the second case, of clinical histories. The annotation schema and guidelines were developed iteratively following the MAMA cycle (Model-Annotate-Model-Annotate) [42]. Five annotation rounds were performed. In each of them, a portion of the records were annotated by more than one annotator to compute inter-annotator agreement metrics. The first two rounds, with 160 records per annotator and 40% overlap, were used to refine annotation criteria and then they were re-annotated to correct errors. In the last three rounds, each annotator annotated 1869 records with a 10% overlap. Annotations were done with Label-Studio tool.[5]

Table 1
Label distribution of the annotated dataset.

Label	Number	Percentage
no-COVID	2,360	47.24%
COVID-confirmed	1,374	27.50%
COVID-suspected	1,125	22.52%
COVID-rejected	60	1.20%
insufficient information	77	1.54%
Total number of reports	4,996	100%

The level of agreement between annotators increased through the subsequent annotation stages. The main annotation disagreements were between COVID-suspected and no-COVID cases. In the last stage, we obtained an inter-annotator agreement (IAA) of 0.798, measured as an average of the Kappa coefficient [43] between pairs of annotators. This IAA indicates good reproducibility of annotations.

To resolve annotation disagreements, the epidemiology residents examined each discrepancy and selected the most appropriate option or made the necessary corrections.

The obtained labeled dataset contains 4996 reports. The number and percentage of reports per label can be seen in Table 1.

3.3. Approaches to automatic classification

We explored different types of machine learning classifiers for the task, combined with different feature sets. As input features, we used the raw text of the reports for the deep learning approaches. For the simpler models (Naïve Bayes, Logistic Regression, and XGBoost), we represented the text as bag of n-grams. The output of the classifier is a label with five possible values: COVID-suspected, COVID-confirmed, COVID-rejected, no-COVID, and insufficient information.

As there exists additional information other than the text reports in the clinical records (i.e., medical specialty, assigned ICD-10 codes when present, and the presence of COVID-related keywords like anosmia), we performed an additional experiment to evaluate whether the addition of this extra information could improve the results or, on the contrary, our algorithms are already performing at their best when based on the textual information alone. This set of additional features is called "extra" in the rest of the paper and this experiment works as a positive control, that is an experiment where a good result is expected. See Appendix A.2 for details.

We trained Naïve Bayes, Logistic Regression and Gradient Boosting Classifier (XGBoost), a bidirectional recurrent neural network (BiLSTM), and two transformer-based approaches: BETO Clínico, and RoBERTa Clínico.

For Naïve Bayes, Logistic Regression, and XGBoost we used their scikit-learn implementations [44]. For the models described below we used a subset of our corpus consisting of approximately 1 million non-annotated clinical notes. From now on we will refer to this corpus as *RawCRC* (Raw Clinical Reports Corpus).

[5] Label Studio: https://labelstud.io/. Last accessed 10/2024.

For the BiLSTM model, we evaluated an implementation without word embeddings and with three different word embeddings to represent the text:

1. general pre-trained embeddings of DCC UChile,[6]
2. domain pre-trained embeddings consisting on SciELO database [45] and Wikipedia articles related to Health domain,[7] and
3. word embeddings trained with the RawCRC corpus.

BETO Clínico and *RoBERTa Clínico* are two models developed by us through further pre-training of previously existing models using the RawCRC corpus. BETO, a Spanish adaptation of BERT [14], and RoBERTa had been pre-trained by their respective developers on different general corpora [13,46]. We started with BETO and roberta-base-biomedical-clinical-es [47], a RoBERTa model further pre-trained on a biomedical corpus in Spanish (which contains a variety of medical text types, but does not include reports similar to ours). We then performed further pre-training on BETO and roberta-base-biomedical-clinical-es using the RawCRC corpus. Finally, we fine-tuned the models using our annotated dataset, as described in the next subsection.

3.4. Experimental settings

Using stratified random sampling, we partitioned the manually annotated corpus into three subsets: training (70%), validation (15%), and testing (15%, i.e., 750 reports). The testing set was held out and not used until the testing phase. Different combinations of hyperparameters for the classifiers were explored, with training performed on the training set and evaluation on the validation set. For each classifier, the hyperparameters that yielded the highest F1 score (harmonic mean between precision and recall) on the validation set were selected. Once the optimal hyperparameters were chosen, due to the limited size of our dataset, the training and validation sets were combined to train the classifiers, which were then tested using the testing set. The BiLSTM models were trained for a maximum of 20 epochs, using a patience parameter of 4 iterations, which interrupts the training if no improvements in performance metrics are observed. In all cases, the classifiers stopped training at less than 8 epochs.

Due to the hardware constraints (we used an Intel® Xeon® E5-2680 v2 CPU with 10 cores and 65 GB of RAM, rather than a GPU), computational trade-offs were critical in our experiments, directly impacting the duration of our experiments and the number of epochs we could afford for further pre-training and fine-tuning of the transformer-based models.

The further pre-training of BETO Clínico and RoBERTa Clínico was done using Whole Word Masking.[8] Despite computational limitations (pre-training these models took approximately 10 days each) restricting us to 3 training epochs, the results were promising, especially for the model resulting from performing TAPT (Task-Adaptive Pretraining) [48] on BETO, which achieved the best results.

Each transformer fine-tuning (performed for 5 epochs with a batch size of 8) took over 3 h, whereas the simplest models were trained in minutes. This disparity made hyperparameter exploration feasible only for the simple models and BiLSTM. In contrast, for transformers, we adopted established configurations from widely used benchmarks [49] (learning rate of 2×10^{-5} and weight decay of 0.01), as exhaustive grid search was prohibitively costly. In line with this, recent work supports that transformers are robust to moderate hyperparameter variations under resource constraints [50].

In order to optimize the performance of our models, we decided to consistently use the ADAM (Adaptive Moment Estimation) [51] optimizer (for BiLSTM) and its variant AdamW[9] (for transformer-based models) for weight updates of the different models, given its proven effectiveness in various deep learning contexts [14,52,53].

The further pre-training of BETO Clínico and RoBERTa Clínico with the RawCRC corpus and the posterior training with our annotated data were done with the HuggingFace transformers library.

Different approaches were compared based on micro-averaged F1. Micro-averaged precision and recall are also informed. The use of other metrics was discarded for several reasons, including dataset imbalance and the consensus within the field regarding the reliability of these metrics [54–56]. All metrics are reported with their 95% confidence intervals, obtained by bootstrapping with 1000 samples with replacement of the original size.

For the best-performing models among both simple and complex approaches, we compared the time series Pearson correlation between their results on the test set and the SNVS 2.0 data for the capital of La Rioja province, for each epidemiological week of 2020.

3.5. Hypothesis testing

In order to statistically test our main and secondary hypotheses (i.e., 1: whether the algorithms perform differently, and 2: whether they perform at their best based on the textual information alone or if the addition of "extra" information improves performance) we run a fixed-effects ANOVA model for each. See Appendix A.1 for details.

4. Results

In this section, we show the results of our ML models for the testing dataset in terms of micro-averaged precision, recall, and F1, we refer to the results shown in the Appendix regarding the performance of adding extra information to the texts, and we present results for each classification class for the best performing of our simpler models (logistic regression) and for the best-performing model overall (BETO Clínico). We also present the confusion matrix of BETO Clínico. We then compare the time series Pearson correlation of Logistic Regression and BETO Clínico against SNVS 2.0 from the capital of the province of La Rioja for confirmed and suspected cases and inform the Pearson correlation between SNVS 2.0 data, and LR, Beto Clínico and U071 (ICD-10 classification for confirmed cases), and the Pearson correlation among LR and BETO Clínico and ICD-10 classification codes for confirmed and suspected cases (U071 and U072). Finally, we analyze the results.

In Table 2 we show the micro-averaged precision, recall and F1 of our models for the testing dataset. In particular, the BiLSTM model performed better without using word embeddings compared to when any of the three different embeddings were used. The best results among the embeddings were achieved with word embeddings trained on our RawCRC corpus. For BiLSTM we show the results without embeddings and with embeddings obtained from our RawCRC corpus.

All predictive models obtain a micro-averaged F1 performance above 79%. Not all models performed equally well, however—transformer-based models BETO Clínico and RoBERTa Clínico did particularly well, whereas Naïve Bayes performed worst. This observation is quantitatively supported by an ANOVA with a significant effect of Model (see Appendix A.1). Interestingly, Logistic Regression fared better than more complex gradient boosting and long short-term memory models (BiLSTM): Logistic Regression F1 performance was above BiLSTM without WE by around 1.17 percentage points, while BiLSTM without WE mean F1 value in turn was higher than XGBoost by 1.42 percentage points. See Supplementary Table A.1 to see

[6] Embeddings for Spanish by Jorge Pérez Rojas: https://github.com/dccuchile/spanish-word-embeddings

[7] https://zenodo.org/records/3626806

[8] When training models for masked language modeling, whole word masking masks whole words instead of individual tokens.

[9] AdamW is an extension of Adam that includes a form of weight regularization to improve model generalization.

Table 2
Micro-averaged precision, recall, and F1 percentage scores achieved by our models for the automatic classification of clinical notes on the test set (ordered by F1 value). Mean across samples ± confidence intervals obtained with bootstrapping with 1000 samples are shown (see Experimental Settings).

Model	Precision	Recall	F1
BETO Clínico	88.08 ± 0.0008	88.18 ± 0.0008	88.13 ± 0.0008
RoBERTa Clínico	86.44 ± 0.0008	87.60 ± 0.0008	87.01 ± 0.0008
Logistic Regression	84.95 ± 0.0009	85.23 ± 0.0008	85.09 ± 0.0008
BiLSTM without WE	83.75 ± 0.0008	84.10 ± 0.0008	83.92 ± 0.0008
BiLSTM with WE	82.70 ± 0.0008	84.02 ± 0.0008	83.35 ± 0.0008
XGBoost	82.31 ± 0.0008	82.70 ± 0.0005	82.50 ± 0.0008
Naïve Bayes	78.58 ± 0.0009	80.20 ± 0.0008	79.38 ± 0.0008

Table 3
Precision (P), Recall (R), and F1 scores for each class for the logistic regression and BETO Clínico models. Support (number of cases) is also shown.

	Logistic Regression			BETO Clínico			
Label	P	R	F1	P	R	F1	Support
COVID-confirmed	88.83	81.07	84.77	90.87	91.75	91.30	206
COVID-rejected	75.00	33.33	46.15	41.18	77.78	53.85	9
COVID-suspected	80.00	80.47	80.24	82.63	81.66	82.14	169
insufficient information	100.00	08.33	15.38	25.00	08.33	12.50	12
no-COVID	86.05	94.07	89.88	92.37	92.37	92.37	354
Weighted average	85.54	85.33	84.59	88.07	88.27	88.03	750

all F1 differences between consecutive models. Furthermore, Logistic Regression had a surprisingly good performance as its F1 was below RoBERTa Clínico by just 1.92 percentage points. Incorporating word embeddings negatively impacted the performance of the BiLSTM, albeit only slightly by less than 1 percentage point.

In order to test whether models performed at their best when using only the textual information from the clinical notes (results shown above), we also trained the models with "Extra" information provided by ICD-10 codes, medical specialty, and explicit identification of symptoms—that is, a positive control by using all available information (i.e., where a good result is expected). Results are shown in Supplementary Fig. A.1. We found that the Extra information contributed only slightly to improve the performance of the more complex classifiers (either transformer-based, gradient boosting, or LSTM) by 1 to 3 percentage points. Logistic Regression was less favored, and Náive Bayes was actually harmed by Extra (see Supplementary Table A.2). This result means that models were actually near top performance without the extra information (see Appendix A.2 for a two-way ANOVA with a significant effect of Extra and significant interaction Model x Extra).

4.1. Analysis of results

From these results, we can conclude that very simple classifiers like Logistic Regression can obtain a satisfactory performance (85.09% F1), even better than BiLSTM. The effort and expertise required to deploy more complex machine learning approaches like transformers-based models, needs to be considered in the light of the 3 percent points of improvement in F1 in the best case (BETO Clínico).

Next, we analyze the performance of the best of the simplest models (Logistic Regression) and the best of the most complex models—which also achieved the best overall performance, BETO Clínico. Table 3 shows the Precision, Recall, F1 scores, and support for each class for both the Logistic Regression and BETO Clínico models.

As can be seen in Table 3, BETO Clínico outperforms Logistic Regression in F1, Recall, and Precision in almost all cases. The exceptions are the class *Insufficient information* where both models perform equally across the three metrics with a support of 12 cases; *No-COVID*, where logistic regression performs better in recall than BETO; and *COVID rejected*, where logistic regression outperforms BETO Clínico with a support of 9 cases. Notably, BETO Clínico outperforms by 11 percentage points in recall for *Confirmed COVID*.

The confusion matrix for BETO Clínico (Fig. 1) shows that, among the three classes with the largest support, COVID-suspected is the most misclassified case, either in relative or absolute terms, with most frequent incorrect labels COVID-confirmed and no-COVID. A large fraction of "Insufficient information" cases are also mislabeled as COVID-suspected. This reflects the intrinsic difficulty in classifying borderline cases.

We can see in Table 3 and Fig. 1 that both the BETO Clínico and logistic regression approaches effectively predict the two majority classes, no-COVID and COVID-confirmed. Logistic Regression has a bias towards amplifying the majority class, leading to the misclassification of some cases as no-COVID. Conversely, both approaches characterize minority classes poorly because they have very few training instances. It is important to note that they were peripheral to the objective of the project and were included only to label unclear cases.

4.2. Comparison with SNVS 2.0 data

We further analyzed the following two models: Logistic Regression (best performance among the simple models) and BETO Clínico (best performance among the complex models). We computed the Pearson correlation between each of the time series generated by these models on the test dataset and the SNVS 2.0 data from the capital of the province of La Rioja for each epidemiological week of 2020 (from Sunday to Saturday).

Fig. 2 shows the comparison of COVID-positive cases provided by our models (Logistic Regression and BETO Clínico), SNVS data for the capital of the province of La Rioja, and U071 classification (confirmed-COVID cases according to ICD-10). In order to compare the temporal features among the time series, the number of cases is normalized to the maximum of each curve (see the non-normalized plot in Supplementary Fig. B.1 in Appendix B). The time series were built for the 750 cases in the testing data set, which has 38 cases with ICD-10 equal to null, 171 COVID-confirmed (U071 code), and 201 COVID-suspected (U072 code). As the SNVS informs very few COVID-suspected cases we plotted confirmed cases only.

As observed in Fig. 2, the algorithms (represented by the blue and orange curves, corresponding to Logistic Regression and BETO Clínico) start detecting positive cases earlier than the U071 code (green curve). The curve shapes are similar in all cases, which validates that the results are comparable to those of the SNVS and U071. However, towards the end of the series, the reported cases seem to decrease. This is possibly due to delays in case reporting (with a minimum delay of a couple of days, which can extend up to a week). The algorithms provide real-time information that would help us assess whether the curve is truly declining.

In Fig. 2 we observe that the time series of BETO and Logistic Regression are more similar to the SNVS 2.0 time series than ICD-10 U071. This is supported by a higher Pearson correlation between any of our models and the SNVS data than between the U071 data and SNVS (see Table 4). This suggests that it is beneficial to employ these algorithms for automatic classification, rather than manually assigning ICD-10 codes. Although we had to train the algorithms based on manual annotations, they were significantly fewer than what would be required to classify all medical records in ICD-10 manually. Regarding the necessity for annotations to train the algorithms, we observed that the annotations do not align with the ICD-10 codes, which justifies the need for making these annotations.

As can be seen in Table 4, data classified as COVID-confirmed using Logistic Regression and BETO Clínico have a higher correlation with the official COVID-19 case counts provided by the SNVS compared to the correlation observed with ICD-10 U071 code (0.840, 0.873, and

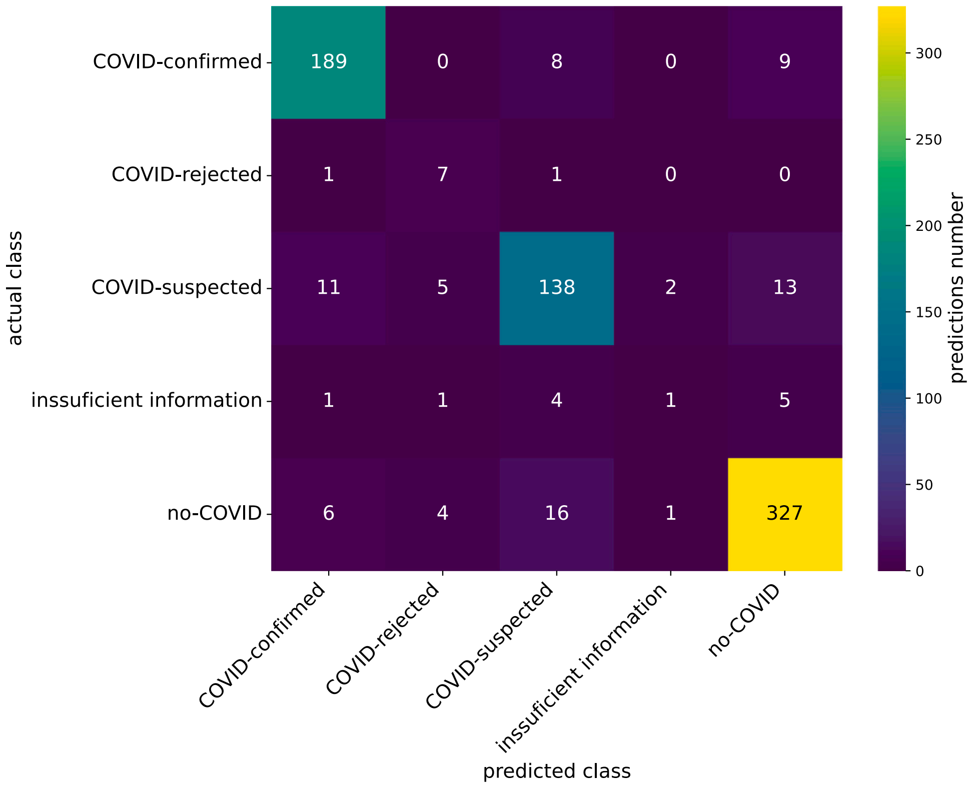

Fig. 1. Confusion matrix of BETO Clínico.

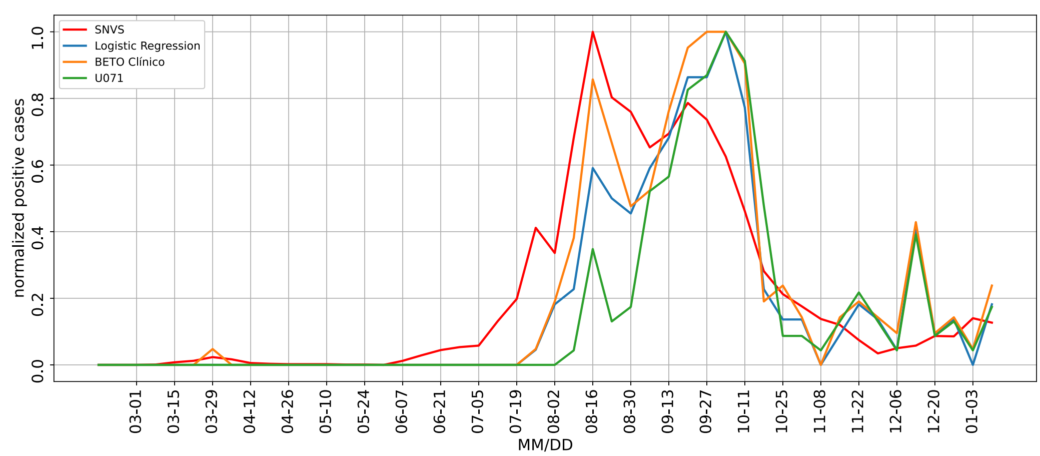

Fig. 2. Time series of COVID positive cases provided by our models (logistic regression and BETO Clínico), SNVS 2.0 for the capital of the province of La Rioja, and U071 classification (confirmed-COVID) in the test dataset. Values are normalized by the maximum values of each time series. Results are shown for each epidemiological week of 2020 (from Sunday to Saturday).

0.663, respectively). BETO Clínico shows a slightly higher correlation with SNVS 2.0 data than Logistic Regression.

We performed a similar analysis with the COVID-suspected cases. In Fig. 3, we present the time series of the results obtained by BETO Clínico and Logistic Regression algorithms, the ICD-10 codes for suspected cases (U072). For comparison, we also plotted the confirmed cases from the SNVS data as above. We can observe that, although both confirmed and suspected SNVS cases peak at around the same time (August 16th), the prediction of suspected cases from our algorithms starts growing around two weeks before than the prediction of confirmed cases (July 5th vs July 19th, respectively). This points to the usefulness of predicting suspected cases in order to anticipate an outbreak.

Table 5 shows the Pearson correlation between the predictions for confirmed and suspected cases from each algorithm (Logistic Regression and BETO Clínico) and the U071 and U072 codes from ICD-10. Both models have a high correlation with the corresponding confirmed/suspected ICD-10 codes (confirmed 0.940 and 0.902, respectively; suspected 0.960 and 0.954, respectively. p-value < 0.05 in all cases.).

5. Discussion

Our study shows that simple classification algorithms enable earlier detection of cases compared to traditional surveillance methods. Our prediction of confirmed cases identifies an emerging trend at an earlier stage than the corresponding ICD-10 code, and our prediction of suspected cases closely follow the real number of suspected cases around

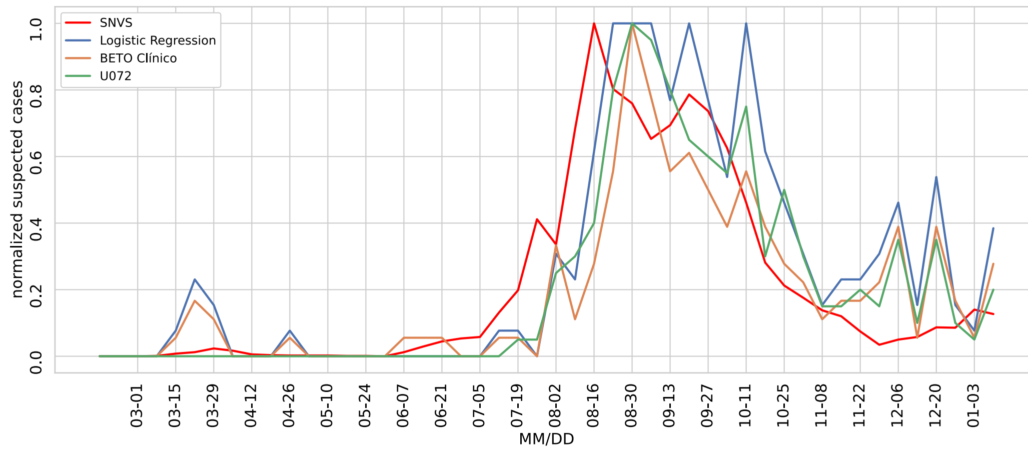

Fig. 3. Time series of COVID-suspected cases provided by our models (Logistic Regression and BETO Clínico) and U072 classification (COVID-suspected) in the test dataset. We also plotted confirmed cases from SNVS 2.0 data for the capital of the province of La Rioja for comparison. Values are normalized by the maximum values of each time series. Results are shown for each epidemiological week of 2020 (from Sunday to Saturday).

Table 4
Pearson correlation and its *p*-value, between SNVS 2.0 data and Logistic Regression, BETO Clínico, and the U071 code from ICD-10 for COVID positive cases in the test dataset (corresponding to time series shown in Fig. 2).

	SNVS 2.0 data	
	Pearson correlation	*p*-value
logistic regression	0.840	8.1×10^{-14}
Beto Clínico	0.873	6.2×10^{-16}
U071	0.663	2.8×10^{-7}

Table 5
Pearson correlation and its *p*-value among the Logistic regression algorithm, and BETO Clínico algorithm, and the U071 (confirmed cases) and U072 codes from ICD-10 (suspected cases).

	Logistic Regression		BETO Clínico	
	Pearson correlation	*p*-value	Pearson correlation	*p*-value
U071 (confirmed)	0.940	5.1×10^{-23}	0.902	1.7×10^{-18}
U072 (suspected)	0.960	3.5×10^{-27}	0.954	7.1×10^{-26}

two weeks ahead in time than the confirmed cases. Furthermore, the fact that our two very different main algorithms (BETO Clínico and Logistic Regression) lead to very similar time evolutions of the number of cases — and very close to the true number according to SNVS data — lends strong support to the reliability and validity of the methodological approach employed in this research. This has important implications for public health response, allowing for timely implementation of effective measures, especially in the context of rapidly emerging epidemics like COVID-19, dengue, and Oropouche fever.

This approach could also prove valuable in retrospective classification settings, such as measles epidemiological surveillance, where proactive institutional and community searches are undertaken to identify previously undetected cases among contacts.

Another aspect to highlight is the importance of real-time information when interpreting the most recent data on an outbreak. The observed increase in cases may not accurately reflect the true trajectory of the outbreak, as there is often a lag between the occurrence of an event and its official reporting. This delay poses a challenge to detecting an outbreak. Real-time data serves as a valuable complement to indicator-based surveillance in public health decision-making.

In addition, our result regarding the small improvement in performance when *extra information* is included tells us that the textual information of EHRs is sufficient by itself to determine with high reliability whether a positive case is being described. Relying on text alone allows for better portability and interoperability of the solution. Furthermore, this suggests that using an automatic classifier could be more effective than assigning a specialist to manually code medical records, thereby allowing these specialists to engage in more useful and interesting tasks.

Cost-effective solutions in resource-constrained settings

As Hossain et al. mention [6], while deep learning algorithms have been successful for NLP tasks, their application in the biomedical field remains challenging. Unlike classic machine learning models, which are often used for health records, deep learning models face disadvantages such as data availability, the complexity of domain-specific textual data, and interpretability. Notably, deep learning-based algorithms require large amounts of data, expensive GPUs, and hundreds of workstations to outperform other methods.

These requirements are critical in resource-constrained settings. Machine learning approaches to epidemiological surveillance of infectious diseases have surged in the last decade [57]. Obstacles and issues, however, have only begun to be noticed and are particularly detrimental in low-resource settings. Most of the implemented solutions are only accessible in developed countries due, for instance, to the need for a large volume of good quality data and trained personnel with available time for research that is typically lacking in developing countries due to the lack of investment in research. When discussing the limits of artificial intelligence in epidemiological surveillance, little attention is paid to the scarce availability of high-cost solutions in resource-limited settings [57]. Many recent reviews point to the difficulties of implementing state-of-the-art machine learning solutions in developing countries including limitations in computational power (in our case it took us 10 days to just pre-train each of our two transformer-based models), yet descriptions of actual implementations of cost-efficient solutions are lacking [57,58]. In addition, most of the published works comparing algorithm performance show marginal increases in the order of a few tenths in F1 values without conducting a trade-off analysis between these slight performance gains and the energetic cost and personnel requirement. These gains often come from switching to energy-hungry, state-of-the-art transformer-based solutions. This purely incremental race for ever higher metrics places developing countries at a disadvantage.

Comparison to other works and data availability

We faced many of the challenges described by other authors of using NLP to classify or extract information from EHRs to improve disease surveillance systems: the concerns around data privacy [3] and the need of data anonymization [6], the multi-disciplinary work needed to address these solutions [3,4], the scarcity of publicly available annotated data (absence in our case) and the high cost of annotating it [5,8], and the highly imbalanced data [4,5]. Because of the right to data privacy, as in other projects, data had to be anonymized. We had to create an annotated data set, which was a costly process, mainly due to the time and knowledge required to create the annotation criteria, the need of finding trained personnel to do the annotations, and the iterations needed until establishing the final annotation criteria. The size of our annotated corpus (4996 clinical reports annotated for classification into COVID-related categories) is bigger than others (1472 notes [20], and 369 notes [26]). Regretfully, as it is frequent in this area of research, due to the sensitivity of the data we are not allowed to publish our corpus, and other Spanish corpora are not publicly available. Therefore, we cannot compare our results to those of other researchers. Besides, some studies use different metrics to publish their results. Nevertheless, we can say that (with other datasets, other language and not exactly the same task) our best results (a micro-averaged F1 of 88.13%) are similar to the rule-based results of [26], that, other than us, used symptom information (micro-averaged F1: 87.6% -and 65.5% with deep learning-) and are similar to [20] with 71%F1 for COVID-assertion. They are also similar to [22], which work on ICD-9 classification, but for different problems.

The sensitivity of the data, the scarcity of publicly annotated datasets, and the cost of annotating data derive in few researches evaluating external validity or using multi-institution data (see [21]). Our research is no exception to this situation. Unlike other works conducted for ICD-9 code classification, we focus on ML techniques and use newer approaches like FastText and BERT (see [6]). Our comparison in terms of the advantages and disadvantages of machine learning models yields similar results to those mentioned by Hossain et al. [6]. Similar to them, In our case, Naïve Bayes, logistic regression and XGBoost were easy and fast to implement, and Naïve Bayes had a lower F1 than logistic regression. We also found BiLSTM computationally expensive, but with worse results than logistic regression. BETO Clínico and RoBERTa Clínico had the best results in terms of F1, but are more resource-intensive in terms of knowledge required to implement them, hardware requirement, and energy consumption.

Rich interdisciplinary nature of the team

The rich interdisciplinary nature of our team (composed of epidemiologists, an anthropologist, a physicist, computer scientists, and a linguist) made the discussions very rich, although challenging. It is remarkable that at the initial stage of the project the team was almost exclusively composed by women, which is usually not the norm.

Adaptation and generalization potential

To address the generalizability and reusability of our methods and research, our annotation criteria can be made available under request.

To adapt our approach to other languages, it would be necessary to compile and annotate new datasets, as well as fine-tune the models accordingly. Our annotation guidelines could serve as a basis for this process, but lexical and grammatical differences should be considered when using them. Therefore, the names of symptoms used to classify EHRs into COVID-related categories should be adjusted to reflect the particularities of each variant or language, as mentioned in [59]. Additionally, based on our results, we recommend performing task-adaptive pretraining using unannotated corpora that matches the target task and language variety. When applying our methods to other variants of Spanish, which may differ lexically and syntactically, a similar approach should be followed, given that our dataset cannot be publicly shared. If the dataset were available, at least one round of annotation would be necessary to assess the effectiveness of our algorithms on a different dataset. Based on the results, it should be analyzed whether further pre-training with unannotated data is needed. Moreover, ensuring that the corpora are representative of the target variety is crucial for successful adaptation.

The main limitations to adapting our methods to other settings are resource availability (obtaining and annotating EHRs) and the computational costs associated with further pre-training models with unannotated data.

When applying our methods to other data, it would be important to analyze the proportion of records in each class, as this could influence model performance, fairness, and generalization capacity. If handling larger datasets, additional computational resources may be required, particularly during the pre-training and fine-tuning phases of transformer-based models. Simpler methods, on the other hand, would not require as much computational power and could potentially demonstrate even greater advantages in such cases.

Concerning changes in medical protocols, work could be done on the automatic detection of symptoms (performed via named entity recognition). We have also worked in this area. This is a more labor-intensive task than phenotype detection (which we conducted through classification). The former strategy is more flexible than phenotype detection, especially with evolving definitions and emerging diseases, and is better suited for syndromic surveillance, as it is adaptable to early signal detection (e.g., monitoring fever). These developments could help address current challenges in epidemiological surveillance, such as detecting and monitoring emerging diseases before new codes are created and implemented (e.g., Oropouche fever) and identifying signals of re-emerging diseases (e.g., measles). Additionally, mapping symptoms into SNOMED CT terms,[10] for example, could facilitate interoperability and standardization by assigning a unique code to a medical concept, regardless of how it is expressed in natural language, which could indirectly help in handling different lexical variants or languages [60].

6. Conclusion

We present state-of-the-art results in the classification of unstructured Spanish EHR texts into COVID-related categories during an emergency scenario, despite limited resources and the absence of ICD-10 codes for the disease at the start of the project. To address these challenges, we developed an annotation criteria and annotated a corpus of 4996 clinical notes, achieving an inter-annotator agreement of 0.798. We then implemented several machine learning algorithms to classify the EHRs. Our initial development was conducted with minimal resources over a three-month period, followed by an additional three months dedicated to comparisons with BETO Clínico and RoBERTa Clínico, along with statistical analyses.

The best performance was achieved by BETO Clínico, a transformer model retrained on an unannotated subset of our dataset, which achieved an F1 score of 88.13% on the test dataset. Surprisingly, Logistic Regression — a much simpler method — scored 85.09% F1. Moreover, logistic regression outperformed the BiLSTM model (F1: 83.92%). Furthermore, data identified as COVID-confirmed through BETO Clínico and Logistic Regression models show a high Pearson correlation with official COVID-19 case counts from the National Health Surveillance System (SNVS 2.0) for the capital of La Rioja. This indicates that simpler models can compete effectively with more complex, resource-intensive approaches like transformer-based models.

[10] Systematized Nomenclature of Medicine – Clinical Terms, https://www.nlm.nih.gov/healthit/snomedct/index.html

The finding that a simpler method achieved only a 3-point difference in F1 score compared to the transformer pre-trained with domain-specific data has significant implications for managing potential future epidemics in low-resource settings. In such contexts, limitations in hardware, available data, and available trained personnel can pose challenges to deploying complex models in urgent scenarios. Therefore, low-resource deployment strategies are highly valuable in public health systems, particularly during the rapid onset of a pandemic or epidemic. In conclusion, our study offers rapid, cost-effective solutions demonstrated through a COVID-19 case study, which can be applied in emergency scenarios in low-resource settings. These approaches enhance automated surveillance and epidemic intelligence, not only for this outbreak but also for addressing similar challenges in future epidemics.

Despite the strengths of this study, certain limitations must be acknowledged. One potential limitation is that the dataset is restricted to a single province in Argentina, which may impact the generalizability of the results. However, the fact that BETO Clínico and Logistic Regression exhibit qualitatively comparable behavior suggests that these models would likely perform similarly when applied to different languages or datasets. Moreover, given the scarcity of computational resources, we were unable to perform 5-fold cross-validation, which would have been the ideal approach for hyperparameter tuning and model robustness evaluation. Instead, we opted for a single validation split to select the hyperparameters and, due to the limited data available, made the practical decision to train the models by combining the training and validation sets. While this strategy maximizes the use of the available data for training, it may introduce certain issues, such as overfitting to the selected hyperparameters, reduced evaluation robustness, and underestimation of metric variability. We, nevertheless, evaluated the models with a completely independent test set. In addition, to address the possible limitations, we applied bootstrapping to assess metric variability. Furthermore, although we emphasize the effectiveness of simpler methods compared to transformers, we must clarify that our capacity to continue pre-training models with unlabeled EHRs was constrained by both the size of the corpus and the available computing power. Our limitations in computational capacity led us to make specific trade-offs: as described above, to reduce computational costs in BiLSTMs and transformers, we opted for a single validation split instead of cross-validation, acknowledging the potential compromise in the robustness of the evaluation metrics. While the lack of advanced computational resources (such as GPUs) limits the number of experiments we can perform and may result in less optimal outcomes, it primarily impacted our experiments with transformer-based models. This constraint limited the number of epochs we could use to train the models and prevented us from conducting an exhaustive grid search. Furthermore, the transformer-based models we implemented, where chosen because they represented the state of the art in clinical NLP at the time the experiments were conducted. However, it is important to notice that our computational limitations would have posed a barrier to implementing larger language models. Moreover, it is important to acknowledge the significant energy costs associated with training large models on advanced hardware [11,12]. These costs, when compared to the annual energy consumption of a country — especially in low-resource settings — highlight the importance of prioritizing sustainable and accessible research practices.

In a real-world, low-resource setting, we would give strong consideration to implementing logistic regression, as it requires considerably less computational power for both training and inference. Additionally, logistic regression is generally easier to implement, with greater availability of human resources compared to transformer-based models. In this case, it also offers comparable performance to more computationally expensive algorithms.

Finally, due to the sensitivity of the data, we are unable to publish our annotated corpus. This limitation, combined with the lack of additional Spanish corpora for this problem, hinders both our capacity and that of other researchers to conduct benchmark comparisons between our work and others. This challenge is frequently encountered in this area of research.

We are currently finalizing our efforts on a potential improvement, focusing on symptom detection. Although this approach involves a more time-consuming implementation phase and is more costly, it has the potential to yield better results given the dynamic nature of diseases. By automating symptom detection, we could also identify outbreaks of different emerging diseases. This solution requires the deployment of a different NLP task, namely named entity recognition rather than classification.

CRediT authorship contribution statement

Javier Petri: Writing – review & editing, Visualization, Software, Methodology, Investigation, Data curation. **Pilar Barcena Barbeira:** Writing – review & editing, Resources, Data curation. **Martina Pesce:** Writing – review & editing, Resources, Data curation. **Verónica Xhardez:** Resources, Funding acquisition. **Rodrigo Laje:** Writing – review & editing, Visualization, Software, Formal analysis. **Viviana Cotik:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Formal analysis, Data curation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Writefull for Overleaf and ChatGPT in order to improve language and readability. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Hypothesis testing

In this appendix, we show the statistical models, results, and effect sizes for the testing of our two hypotheses:

1. Whether the models perform differently.
2. Whether adding "Extra" information improves performance.

Table A.1
Consecutive differences from the list of model results (see Table 2). All p-values are corrected (Sidak correction for 6 estimates).

Models	F1 diff estimate	95% CI	t ratio (df=6993)	p value
BiLSTM with WE vs. Bayes	0.0089	[0.0073; 0.0105]	14.28	0.0001
BiLSTM without WE vs. BiLSTM with WE	0.0117	[0.0101; 0.0134]	18.86	0.0001
XGBoost vs. BiLSTM without WE	0.0085	[0.0068; 0.0101]	13.59	0.0001
LogReg vs. XGBoost	0.0242	[0.0225; 0.0258]	38.77	0.0001
RoBERTa Clínico vs. LogReg	0.0232	[0.0215; 0.0248]	37.15	0.0001
BETO Clínico vs. RoBERTa Clínico	0.0127	[0.0111; 0.0144]	20.45	0.0001

A.1. Model performance

We tested whether model results shown in Table 2 differed significantly by performing a one-way ANOVA with F1 as dependent variable and Model as factor (levels: Bayes/BiLSTM with WE/BiLSTM without WE/XGBoost / LogReg/RoBERTa Clínico/BETO Clínico; input was the set of 1000 bootstrapped F1 values from every model) that yielded a significant effect of Model ($F(6) = 5993$, $p = 2 \times 10^{-16}$) with a rather large effect size ($\eta^2 = 0.84$). All pairwise post-hoc comparisons with Tukey-adjusted p-values were significant (21 comparisons; $p = 0.0001$ after correction). In Table A.1 we show the list of models (ordered by mean F1 value) and the consecutive comparisons only. The largest difference in F1 between consecutive models is around 2 percentage points (LogReg vs XGBoost, and RoBERTa Clínico vs LogReg). We used the R function `aov` for ANOVA, function `eta_squared` from library `effectsize`, and `emmeans` and `contrast` from the `emmeans` library for post-hoc comparisons.

A.2. Effect of adding "Extra" information

We wanted to test whether models performed at their best when based on the textual descriptions or performance could be improved by adding all available information from the HCE (i.e., in statistical terms a "positive control" where a good result is expected). Our rationale was that, if models performed at their best based on the text of the clinical notes only (bag of words and bag of n-grams), then adding further information ("Extra" information: ICE-10 codes, medical specialty, and the presence of relevant symptoms) would lead to either a small or nonsignificant increase in performance. To this, we trained/tested the models as before, this time with Extra information, and included the results in the dataset (see Fig. A.1). We then ran a two-way ANOVA with F1 as independent variable and factors Model (levels Bayes/BiLSTM with WE/BiLSTM without WE/XGBoost / LogReg/RoBERTa Clínico/BETO Clínico) and Extra (levels Yes/No), and an interaction term Model x Extra. Input was the set of 1000 bootstrapped F1 values from every Model/Extra combination. The interaction effect was significant ($F(6) = 435.14$, $p = 2 \times 10^{-16}$, $\eta_p^2 = 0.16$), meaning that some models benefited from the Extra information while others did not, or did in different amounts. We then performed 6 post-hoc comparisons between Yes/No levels of the Extra factor (one comparison for every model; Tukey-adjusted p-values). All models except Bayes benefited from the "extra" information as shown in Table A.2. The size of the effect, however, ranged from small to moderate (most benefited model: BiLSTM with WE by 3 percentage points; least benefited model: LogReg by less than 1 percentage point). We used the R function `aov` for ANOVA, `eta_squared` from library `effectsize`, and `emmeans`, `pairs` and `test` from the `emmeans` library for post-hoc comparisons.

Table A.2
Including Extra information as a positive control (Extra=Yes) to test whether models performed at their best when based on textual descriptions only (Extra=No). Difference in F1 between Yes/No Extra information for every model. All p-values are FDR-corrected for multiple comparisons.

Models	Yes-No diff estimate	95% CI	t ratio (df=13986)	p value
Bayes	−0.0073	[-0.0084; −0.0061]	−11.86	2×10^{-16}
BiLSTM with WE	0.0307	[0.0296; 0.0319]	50.23	2×10^{-16}
BiLSTM w/o WE	0.0161	[0.0149; 0.0173]	26.30	2×10^{-16}
XGBoost	0.0284	[0.0272; 0.0296]	46.45	2×10^{-16}
LogReg	0.0082	[0.0070; 0.0094]	13.44	2×10^{-16}
RoBERTa Clínico	0.0163	[0.0151; 0.0175]	26.58	2×10^{-16}
BETO Clínico	0.0127	[0.0115; 0.0139]	20.69	2×10^{-16}

Appendix B. Absolute number of COVID-confirmed cases

Fig. B.1 shows the time series of absolute number of COVID-confirmed cases. This figure is a companion to Fig. 2.

Appendix C. Potential biases

During the project, a study was conducted by project participants who are not authors of this paper on the potential biases of the predictive algorithms developed and their impact on the results [61]. Nine fairness criteria were considered to detect bias in the classifiers. They were based on the predicted output (group fairness and conditional statistical parity) and on both the predicted and actual output (predictive parity, false positive error rate balance, false negative error rate balance, equalized odds, conditional use accuracy equality, overall accuracy equality, and Treatment equality). Sensitive attributes such as biological sex, transgender status, nationality, age, and race were considered to identify protected groups. The study discusses the criteria, decisions made, and the complexities involved in evaluating bias, as fairness definitions may be incompatible with each other depending on the application context. Due to time constraints, the proposed evaluation methodology could not be implemented.

Although it would be interesting, we were not able to conduct a bias study for the annotated data. For the sake of preserving patients' privacy, we do not have access to the sensitive attributes corresponding to the records, whose textual data we worked with. However, related to the data statement, we can say, as it was mentioned in Section 3.1, that all the annotators are native speakers of Rioplatense Spanish (the variant of Spanish in which the electronic health records are written), and that two annotators are female and one is male. All annotators are in the age range of 20–30 years. So, we expect that no significant bias was introduced in the annotations.

Finally, we would like to add that a study of the entire database, which represents 53.75% of the population of La Rioja province [62], was conducted to analyze the number of transgender patients, age ranges of patients, sex (53.42% female, the rest male), ethnicity (only 0.43% are from indigenous and Afro-descendant communities, the rest are general population), medical coverage (54.54% do not have health insurance, the rest do), and the distribution of patients across multiple attributes (e.g., number of patients by age and sex and by ethnic group and sex). Most age ranges are adequately represented in this database, except for the ages between 0 to 9 years and 10 to 19 years, which are underrepresented, and the age range of 20 to 29 years, which is overrepresented. The proportions represented in the database for Female and Male sex are similar to those of the Argentinian 2010 census [63]. It is estimated that the data regarding races may be

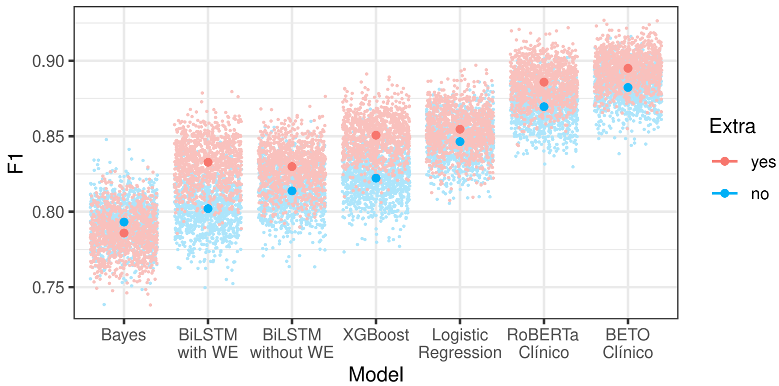

Fig. A.1. F1 as a function of model and Extra information. Small dots represent 1000 bootstrapped samples for every model and condition. Big dots are mean across samples ±95% confidence intervals (indiscernible at this scale). Mean values in the "no Extra" condition correspond to Table A.1.

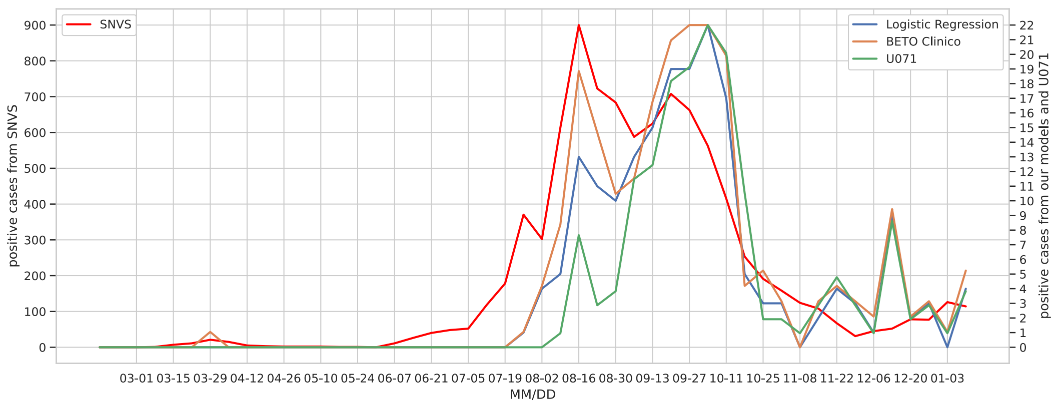

Fig. B.1. Time series of COVID positive cases from SNVS data for the capital of the province of La Rioja (left axis), and our models (Logistic Regression and BETO Clínico) and U071 classification (confirmed-COVID) in the test dataset (right axis).

underrepresented due to the methods of information capture and the comparison of the database values and the 2010 census. As shown in other studies [64], for all specialties, there are consistently more consultations from women than from men, both for scheduled appointments and for emergency care (although in the latter, the difference is not as noticeable). In this study, word cloud analysis was done to identify if there were different characteristics in male and female records, allowing for a quick exploration of the free text fields, which could help detect potential differential patterns in language use across different subsets of the population. To do this, the data was filtered with specific criteria documented in [65], and random and balanced sampling was performed for each sex. Regarding COVID-related data, the frequency of words by type of care was similar for both sexes.

Although the analyses carried out in this study [65] were performed for the entire database and not on the subset of annotated data or the further pre-training data, we assume that the distribution of the annotated data for the different protected attributes is in line with what was observed in the full dataset.

Data availability

Due to the sensitivity of the data and according to the requirements of the sponsor of the project, the generated corpus cannot be made publicly available. The annotation criteria will be publicly available upon publication of this work.

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